CHAPTER 17

SEISMIC DESIGN REQUIREMENTS FOR SEISMICALLY ISOLATED STRUCTURES

17.1 GENERAL

Every seismically isolated structure and every portion thereof shall be designed and constructed in accordance with the requirements of this section and the applicable requirements of this standard.

17.1.1 Definitions. The following definitions apply only to the seismically isolated structure provisions of Chapter 17 and are in addition to the definitions presented in Chapter 11.

BASE LEVEL: The first level of the isolated structure above the isolation interface.

DISPLACEMENT RESTRAINT SYSTEM: A collection of structural elements that limits lateral displacement of seismically isolated structures caused by the maximum considered earthquake.

EFFECTIVE DAMPING: The value of equivalent viscous damping corresponding to energy dissipated during cyclic response of the isolation system.

EFFECTIVE STIFFNESS: The value of the lateral force in the isolation system, or an element thereof, divided by the corresponding lateral displacement.

ISOLATION INTERFACE: The boundary between the upper portion of the structure, which is isolated, and the lower portion of the structure, which moves rigidly with the ground.

ISOLATION SYSTEM: The collection of structural elements that includes all individual isolator units, all structural elements that transfer force between elements of the isolation system, and all connections to other structural elements. The isolation system also includes the wind-restraint system, energy-dissipation devices, and/or the displacement restraint system if such systems and devices are used to meet the design requirements of this chapter.

ISOLATOR UNIT: A horizontally flexible and vertically stiff structural element of the isolation system that permits large lateral deformations under design seismic load. An isolator unit is permitted to be used either as part of, or in addition to, the weight-supporting system of the structure.

MAXIMUM DISPLACEMENT: The maximum lateral displacement, excluding additional displacement caused by actual and accidental torsion, required for design of the isolation system. The maximum displacement is to be computed separately using upper bound and lower bound properties.

SCRAGGING: Cyclic loading or working of rubber products, including elastomeric isolators, to effect a reduction in stiffness properties, a portion of which is recovered over time.

TOTAL MAXIMUM DISPLACEMENT: The total maximum lateral displacement, including additional displacement caused by actual and accidental torsion, required for verification of the stability of the isolation system or elements thereof, design of structure separations, and vertical load testing of isolator unit prototypes. The total maximum displacement is to be computed separately using upper bound and lower bound properties.

WIND-RESTRAINT SYSTEM: The collection of structural elements that provides restraint of the seismically isolated structure for wind loads. The wind-restraint system is permitted to be either an integral part of isolator units or a separate device.

17.1.2 Symbols. Symbols presented in this section apply only to the seismically isolated structure provisions of Chapter 17 and are in addition to the symbols presented in Chapter 11.

b = shortest plan dimension of the structure [ft (mm)], measured perpendicular to d

B_M = numerical coefficient as set forth in Table 17.5-1 for effective damping equal to β_M

C_{vx} = vertical distribution factor

d = longest plan dimension of the structure [ft (mm)], measured perpendicular to b

D_M = maximum displacement [in. (mm)], at the center of rigidity of the isolation system in the direction under consideration, as prescribed by Eq. (17.5-1)

D'_M = maximum displacement [in. (mm)], at the center of rigidity of the isolation system in the direction under consideration, as prescribed by Eq. (17.6-1)

D_{TM} = total maximum displacement [in. (mm)], of an element of the isolation system including both translational displacement at the center of rigidity and the component of torsional displacement in the direction under consideration, as prescribed by Eq. (17.5-3)

e = actual eccentricity [ft (mm)], measured in plan between the center of mass of the structure above the isolation interface and the center of rigidity of the isolation system, plus accidental eccentricity [ft (mm)] taken as 5% of the maximum building dimension perpendicular to the direction of force under consideration

E_{loop} = energy dissipated [kip-in. (kN-mm)], in an isolator unit during a full cycle of reversible load over a test displacement range from Δ^+ to Δ^-, as measured by the area enclosed by the loop of the force-deflection curve

F^+ = maximum positive force [kips (kN)] in an isolator unit during a single cycle of prototype testing at a displacement amplitude of Δ^+

F^- = minimum negative force [kips (kN)] in an isolator unit during a single cycle of prototype testing at a displacement amplitude of Δ^-

F_x = lateral seismic force [kips (kN)] at level x as prescribed by Eq. (17.5-9)

h_i, h_l, h_x = height [ft (m)] above the isolation interface of level i, l, or x

h_{sx} = height of story below level x

k_{eff} = effective stiffness [kip/in (kN/mm)] of an isolator unit, as prescribed by Eq. (17.8-1)

k_M = effective stiffness [kip/in. (kN/mm)] of the isolation system in the horizontal direction under consideration

L = effect of live load in Chapter 17

N = number of isolator units

P_T = ratio of the effective translational period of the isolation system to the effective torsional period of the isolation system, as calculated by dynamic analysis or as prescribed by Eq. 17.5-4 but need not be taken as less than 1.0

r_I = radius of gyration of the isolation system [ft (mm)],

R_I = numerical coefficient related to the type of seismic force-resisting system above the isolation system

T_{fb} = fundamental period [s] of the structure above the isolation interface determined using a modal analysis assuming fixed-base conditions

T_M = effective period [s] of the seismically isolated structure at the displacement D_M in the direction under consideration, as prescribed by Eq. (17.5-2)

V_b = total lateral seismic design force or shear on elements of the isolation system or elements below isolation system [kips (kN)], as prescribed by Eq. (17.5-5)

V_s = total lateral seismic design force or shear on elements above the base level [kips (kN)], as prescribed by Eq. (17.5-6) and the limits of Section 17.5.4.3;

V_{st} = total unreduced lateral seismic design force or shear on elements above the base level [kips (kN)], as prescribed by Eq. (17.5-7)

W = effective seismic weight [kips (kN)], of the structure above the isolation interface, as defined by Section 12.7.2

W_s = effective seismic weight [kips (kN)], of the structure above the isolation interface, as defined by Section 12.7.2, excluding the effective seismic weight [kips (kN)] of the base level

w_i,w_l,w_x = portion of W that is located at or assigned to level i, l, or x [kips (kN)]

x_i,y_i = horizontal distances [ft (mm)] from the center of mass to the ith isolator unit in the two horizontal axes of the isolation system

y = distance [ft (mm)], between the center of rigidity of the isolation system and the element of interest measured perpendicular to the direction of seismic loading under consideration

β_{eff} = effective damping of the isolation system, as prescribed by Eq. (17.8-2)

β_M = effective damping of the isolation system at the displacement D_M, as prescribed by Eq. (17.2-4)

Δ^+ = maximum positive displacement [in (mm)] of an isolator unit during each cycle of prototype testing

Δ^- = minimum negative displacement [in (mm)] of an isolator unit during each cycle of prototype testing

λ_{max} = property modification factor for calculation of the maximum value of the isolator property of interest, used to account for all sources of isolator property variability, as defined in Section 17.2.8.4

λ_{min} = property modification factor for calculation of the minimum value of the isolator property of interest, used to account for all sources of isolator property variability, as defined in Section 17.2.8.4

$\lambda_{(ae,\,max)}$ = property modification factor for calculation of the maximum value of the isolator property of interest, used to account for aging effects and environmental conditions, as defined in Section 17.2.8.4

$\lambda_{(ae,\,min)}$ = property modification factor for calculation of the minimum value of the isolator property of interest, used to account for aging effects and environmental conditions, as defined in Section 17.2.8.4

$\lambda_{(spec,\,max)}$ = property modification factor for calculation of the maximum value of the isolator property of interest, used to account for permissible manufacturing variation on the average properties of a group of same-sized isolators, as defined in Section 17.2.8.4

$\lambda_{(spec,\,min)}$ = property modification factor for calculation of the minimum value of the isolator property of interest, used to account for permissible manufacturing variation on the average properties of a group of same-sized isolators, as defined in Section 17.2.8.4

$\lambda_{(test,\,max)}$ = property modification factor for calculation of the maximum value of the isolator property of interest, used to account for heating, rate of loading, and scragging, as defined in Section 17.2.8.4

$\lambda_{(test,\,min)}$ = property modification factor for calculation of the minimum value of the isolator property of interest, used to account for heating, rate of loading, and scragging, as defined in Section 17.2.8.4

ΣE_M = total energy dissipated [kip-in. (kN-mm)], in the isolation system during a full cycle of response at displacement D_M

$\Sigma|F_D^+|_{max}$ = sum, for all isolator units, of the maximum absolute value of force [kips (kN)], at a positive displacement equal to D_M

$\Sigma|F_D^-|_{max}$ = sum, for all isolator units, of the maximum absolute value of force [kips (kN)], at a negative displacement equal to D_M

17.2 GENERAL DESIGN REQUIREMENTS

17.2.1 Importance Factor. All portions of the structure, including the structure above the isolation system, shall be assigned a risk category in accordance with Table 1.5-1. The Importance Factor, I_e, shall be taken as 1.0 for a seismically isolated structure, regardless of its risk category assignment.

17.2.2 Configuration. Each isolated structure shall be designated as having a structural irregularity if the structural configuration above the isolation system has a Type 1b horizontal structural irregularity, as defined in Table 12.3-1, or Type 1a, 1b, 5a, 5b vertical irregularity, as defined in Table 12.3-2.

17.2.3 Redundancy. A redundancy factor, ρ, shall be assigned to the structure above the isolation system based on requirements of Section 12.3.4. The value of the redundancy factor, ρ, is permitted to be equal to 1.0 for isolated structures that do not have a structural irregularity, as defined in Section 17.2.2.

17.2.4 Isolation System

17.2.4.1 Environmental Conditions. In addition to the requirements for vertical and lateral loads induced by wind and earthquake, the isolation system shall provide for other environmental conditions, including aging effects, creep, fatigue, operating temperature, and exposure to moisture or damaging substances.

17.2.4.2 Wind Forces. Isolated structures shall resist design wind loads at all levels above the isolation interface. At the isolation interface, a wind-restraint system shall be provided to limit lateral displacement in the isolation system to a value equal to that required between floors of the structure above the isolation interface in accordance with Section 17.5.6.

17.2.4.3 Fire Resistance. Fire resistance for the isolation system shall provide at least the same degree of protection as the fire resistance required for the columns, walls, or other such gravity-bearing elements in the same region of the structure.

17.2.4.4 Lateral Restoring Force. The isolation system shall be configured, for both upper bound and lower bound isolation system properties, to produce a restoring force such that the lateral force at the corresponding maximum displacement is at least 0.025 W greater than the lateral force at 50% of the corresponding maximum displacement.

17.2.4.5 Displacement Restraint. The isolation system shall not be configured to include a displacement restraint that limits lateral displacement caused by risk-targeted maximum considered earthquake (MCE_R) ground motions to less than the total maximum displacement, D_{TM}, unless the seismically isolated structure is designed in accordance with all of the following criteria:

1. MCE_R response is calculated in accordance with the dynamic analysis requirements of Section 17.6, explicitly considering the nonlinear characteristics of the isolation system and the structure above the isolation system.
2. The ultimate capacity of the isolation system and structural elements below the isolation system shall exceed the strength and displacement demands of the MCE_R response.
3. The structure above the isolation system is checked for stability and ductility demand of the MCE_R response.
4. The displacement restraint does not become effective at a displacement less than 0.6 times the total maximum displacement.

17.2.4.6 Vertical-Load Stability. Each element of the isolation system shall be designed to be stable under the design vertical load where it is subjected to a horizontal displacement equal to the total maximum displacement. The design vertical load shall be computed using load combination 2 of Section 17.2.7.1 for the maximum vertical load and load combination 3 of Section 17.2.7.1 for the minimum vertical load.

17.2.4.7 Overturning. The factor of safety against global structural overturning at the isolation interface shall not be less than 1.0 for required load combinations. All gravity and seismic loading conditions shall be investigated. Seismic forces for overturning calculations shall be based on MCE_R ground motions, and W shall be used for the vertical restoring force.

Local uplift of individual elements shall not be allowed unless the resulting deflections do not cause overstress or instability of the isolator units or other structure elements.

17.2.4.8 Inspection and Replacement. All of the following items shall be addressed as part of the long-term inspection and replacement program:

1. Access for inspection and replacement of all components of the isolation system shall be provided.
2. A registered design professional (RDP) shall complete a final series of observations of structure separation areas and components that cross the isolation interface before the issuance of the certificate of occupancy for the seismically isolated structure. Such observations shall verify that conditions allow free and unhindered displacement of the structure up to the total maximum displacement and that components that cross the isolation interface have been constructed to accommodate the total maximum displacement.
3. Seismically isolated structures shall have a monitoring, inspection, and maintenance plan for the isolation system established by the registered design professional responsible for the design of the isolation system.
4. Remodeling, repair, or retrofitting at the isolation system interface, including that of components that cross the isolation interface, shall be performed under the direction of a registered design professional.

17.2.4.9 Quality Control. A quality control testing program for isolator units shall be established by the registered design professional responsible for the structural design, incorporating the production testing requirements of Section 17.8.5.

17.2.5 Structural System

17.2.5.1 Horizontal Distribution of Force. A horizontal diaphragm or other structural elements shall provide continuity above the isolation interface and shall have adequate strength and ductility to transmit forces from one part of the structure to another.

17.2.5.2 Minimum Building Separations. Minimum separations between the isolated structure and surrounding retaining walls or other fixed obstructions shall not be less than the total maximum displacement.

17.2.5.3 Nonbuilding Structures. Nonbuilding structures shall be designed and constructed in accordance with the requirements of Chapter 15 using design displacements and forces calculated in accordance with Sections 17.5 or 17.6.

17.2.5.4 Steel Ordinary Concentrically Braced Frames. Steel ordinary concentrically braced frames are permitted as the seismic force-resisting system in seismically isolated structures assigned to Seismic Design Category D, E, and F and are permitted to a height of 160 ft (48.4 m) or less provided that all of the following design requirements are satisfied:

1. The value of R_I as defined in Section 17.5.4 is 1.0.
2. The total maximum displacement (D_{TM}) as defined in Eq. (17.5-3) shall be increased by a factor of 1.2.

17.2.5.5 Isolation System Connections. Moment–resisting connections of structural steel elements of the seismic isolation system below the base level are permitted to conform to the requirements for ordinary steel moment frames of AISC 341, E1.6a and E1.6b.

17.2.6 Elements of Structures and Nonstructural Components. Parts or portions of an isolated structure, permanent nonstructural components and the attachments to them, and the attachments for permanent equipment supported by a structure shall be designed to resist seismic forces and displacements as prescribed by this section and the applicable requirements of Chapter 13.

17.2.6.1 Components at or above the Isolation Interface. Elements of seismically isolated structures and nonstructural components, or portions thereof that are at or above the isolation interface, shall be designed to resist a total lateral seismic force equal to the maximum dynamic response of the element or component under consideration determined using a response history analysis.

EXCEPTION: Elements of seismically isolated structures and nonstructural components or portions designed to resist seismic forces and displacements as prescribed in Chapter 12 or 13 as appropriate are not required to meet this provision.

17.2.6.2 Components Crossing the Isolation Interface. Elements of seismically isolated structures and nonstructural components, or portions thereof that cross the isolation interface, shall be designed to withstand the total maximum displacement and to accommodate on a long-term basis any permanent residual displacement.

17.2.6.3 Components below the Isolation Interface. Elements of seismically isolated structures and nonstructural components, or portions thereof that are below the isolation interface, shall be designed and constructed in accordance with the requirements of Section 12.1 and Chapter 13.

17.2.7 Seismic Load Effects and Load Combinations. All members of the isolated structure, including those not part of the seismic force-resisting system, shall be designed using the seismic load effects of Section 12.4 and the additional load combinations of Section 17.2.7.1 for design of the isolation system and for testing of prototype isolator units.

17.2.7.1 Isolator Unit Vertical Load Combinations. The average, minimum, and maximum vertical load on each isolator unit type shall be computed from application of horizontal seismic forces, Q_E, caused by MCE_R ground motions and the following applicable vertical load combinations:

1. Average vertical load: load corresponding to 1.0 dead load plus 0.5 live load.
2. Maximum vertical load: load combination 6 of Section 2.3.6, where E is given by Eq. (12.4-1) and S_{DS} is replaced by S_{MS} in Eq. (12.4-4a).
3. Minimum vertical load: load combination 7 of Section 2.3.6, where E is given by Eq. (12.4-2) and S_{DS} is replaced by S_{MS} in Eq. (12.4-4a).

17.2.8 Isolation System Properties

17.2.8.1 Isolation System Component Types. All components of the isolation system shall be categorized and grouped in terms of common type and size of isolator unit and common type and size of supplementary damping device, if such devices are also components of the isolation system.

17.2.8.2 Isolator Unit Nominal Properties. Isolator unit nominal design properties shall be based on average properties over the three cycles of prototype testing, specified by Item 2 of Section 17.8.2.2. Variation in isolator unit properties with vertical load are permitted to be established based on a single representative deformation cycle by averaging the properties determined using the three vertical load combinations specified in Section 17.2.7.1, at each displacement level, where required to be considered by Section 17.8.2.2.

EXCEPTION: If the measured values of isolator unit effective stiffness and effective damping for load combination 1 of Section 17.2.7.1 differ by less than 15% from the those based on the average of measured values for the three vertical load combinations of Section 17.2.7.1, then nominal design properties are permitted to be computed only for load combination 1 of Section 17.2.7.1.

17.2.8.3 Bounding Properties of Isolation System Components. Bounding properties of isolation system components shall be developed for each isolation system component type. Bounding properties shall include variation in all of the following component properties:

1. Measured by prototype testing, Item 2 of Section 17.8.2.2, considering variation in prototype isolator unit properties caused by required variation in vertical test load, rate of test loading or velocity effects, effects of heating during cyclic motion, history of loading, scragging (temporary degradation of mechanical properties with repeated cycling), and other potential sources of variation measured by prototype testing,
2. Permitted by manufacturing specification tolerances used to determine acceptability of production isolator units, as required by Section 17.8.5, and
3. Because of aging and environmental effects, including creep, fatigue, contamination, operating temperature and duration of exposure to that temperature, and wear over the life of the structure.

17.2.8.4 Property Modification Factors. Maximum and minimum property modification (λ) factors shall be used to account for variation of the nominal design parameters of each isolator unit type for the effects of heating caused by cyclic dynamic motion, loading rate, scragging and recovery, variability in production bearing properties, temperature, aging, environmental exposure, and contamination. When manufacturer-specific qualification test data in accordance with Section 17.8 have been approved by the registered design professional, these data are permitted to be used to develop the property modification factors, and the maximum and minimum limits of Eqs. (17.2-1) and (17.2-2) need not apply. When qualification test data in accordance with Section 17.8 have not been approved by the registered design professional, the maximum and minimum limits of Eqs. (17.2-1) and (17.2-2) shall apply.

Property modification factors (λ) shall be developed for each isolator unit type, and when applied to the nominal design parameters shall envelop the hysteretic response for the range of demands from $\pm 0.5D_M$ up to and including the maximum displacement, $\pm D_M$. Property modification factors for environmental conditions are permitted to be developed from data that need not satisfy the similarity requirements of Section 17.8.2.7.

For each isolator unit type, the maximum property modification factor, λ_{max}, and the minimum property modification factor, λ_{min}, shall be established from contributing property modification factors in accordance with Eqs. (17.2-1) and (17.2-2), respectively:

$$\lambda_{max} = (1 + (0.75 \times (\lambda_{(ae,\,max)} - 1))) \times \lambda_{(test,\,max)} \times \lambda_{(spec,\,max)} \geq 1.8 \qquad (17.2\text{-}1)$$

$$\lambda_{min} = (1 - (0.75 \times (1 - \lambda_{(ae,\,min)}))) \times \lambda_{(test,\,min)} \times \lambda_{(spec,\,min)} \leq 0.60 \qquad (17.2\text{-}2)$$

where

$\lambda_{(ae,\,max)}$ = property modification factor for calculation of the maximum value of the isolator property of interest, used to account for aging effects and environmental conditions.

$\lambda_{(ae,\,min)}$ = property modification factor for calculation of the minimum value of the isolator property of interest, used to account for aging effects and environmental conditions.

$\lambda_{(test,\,max)}$ = property modification factor for calculation of the maximum value of the isolator property of interest, used to account for heating, rate of loading, and scragging.

$\lambda_{(test, min)}$ = property modification factor for calculation of the minimum value of the isolator property of interest, used to account for heating, rate of loading, and scragging.

$\lambda_{(spec, max)}$ = property modification factor for calculation of the maximum value of the isolator property of interest, used to account for permissible manufacturing variation on the average properties of a group of same-sized isolators.

$\lambda_{(spec, min)}$ = property modification factor for calculation of the minimum value of the isolator property of interest, used to account for permissible manufacturing variation on the average properties of a group of same-sized isolators.

EXCEPTION: If the prototype isolator testing is conducted on a full-scale specimen that satisfies the dynamic test data of Section 17.8.2.3, then the values of the property modification factors shall be based on the test data, and the upper and lower limits of Eqs. (17.2-1) and (17.2-2) need not apply.

17.2.8.5 Upper Bound and Lower Bound Force-Deflection Behavior of Isolation System Components. A mathematical model of upper bound force-deflection (loop) behavior of each type of isolation system component shall be developed. Upper bound force-deflection behavior of isolation system components that are essentially hysteretic devices (e.g., isolator units) shall be modeled using the maximum values of isolator properties calculated using the property modification factors of Section 17.2.8.4. Upper bound force-deflection behavior of isolation system components that are essentially viscous devices (e.g., supplementary viscous dampers) shall be modeled in accordance with the requirements of Chapter 18 for such devices.

A mathematical model of lower bound force-deflection (loop) behavior of each type of isolation system component shall be developed. Lower bound force-deflection behavior of isolation system components that are essentially hysteretic devices (e.g., isolator units) shall be modeled using the minimum values of isolator properties calculated using the property modification factors of Section 17.2.8.4. Lower bound force-deflection behavior of isolation system components that are essentially viscous devices (e.g., supplementary viscous dampers) shall be modeled in accordance with the requirements of Chapter 18 for such devices.

17.2.8.6 Isolation System Properties at Maximum Displacements. The effective stiffness, k_M, of the isolation system at the maximum displacement, D_M, shall be computed using both upper bound and lower bound force-deflection behavior of individual isolator units, in accordance with Eq. (17.2-3):

$$k_M = \frac{\sum |F_M^+| + \sum |F_M^-|}{2D_M} \tag{17.2-3}$$

The effective damping, β_M, of the isolation system at the maximum displacement, D_M, in inches (mm) shall be computed using both upper bound and lower bound force-deflection behavior of individual isolator units, in accordance with Eq. (17.2-4):

$$\beta_M = \frac{\sum E_M}{2\pi k_M D_M^2} \tag{17.2-4}$$

where

$\sum E_M$ = total energy dissipated [kips-in. (kN-mm)] in the isolation system during a full cycle of response at the displacement D_M.

$\sum F_M^+$ = sum, for all isolator units, of the absolute value of force [kips (kN)] at a positive displacement equal to D_M.

$\sum F_M^-$ = sum, for all isolator units, of the absolute value of force [kips (kN)] at a negative displacement equal to D_M.

17.2.8.7 Upper Bound and Lower Bound Isolation System Properties at Maximum Displacement. The analysis of the isolation system and structure shall be performed separately for upper bound and lower bound properties, and the governing case for each response parameter of interest shall be used for design. In addition, the analysis shall comply with all of the following:

1. For the equivalent lateral force procedure, and for the purposes of establishing minimum forces and displacements for dynamic analysis, the following variables shall be calculated independently for upper bound and lower bound isolation system properties: k_M and β_M per Section 17.2.8.6 (Eqs. (17.2-3) and (17.2-4)), D_M per Section 17.5.3.1 (Eq. (17.5-1)), T_M per Section 17.5.3.2 (Eq. (17.5-2)), D_{TM} per Section 17.5.3.3 (Eq. (17.5-3)), V_b per Section 17.5.4.1 (Eq. (17.5-5)), and V_s and V_{st} per Section 17.5.4.2 (Eqs. (17.5-6) and (17.5-7)).
2. The limitations on V_s established in Section 17.5.4.3 shall be evaluated independently for both upper bound and lower bound isolation system properties, and the most adverse requirement shall govern.
3. For the equivalent lateral force procedure and for the purposes of establishing minimum story shear forces for response spectrum analysis, the vertical force distribution from Section 17.5.5 shall be determined separately for upper bound and lower bound isolation system properties. This determination will require independent calculation of F_1, F_x, C_{vx}, and k, per Eqs. (17.5-8) through (17.5-11), respectively.

17.3 SEISMIC GROUND MOTION CRITERIA

17.3.1 Site-Specific Seismic Hazard. The MCE_R response spectrum requirements of Sections 11.4.5 and 11.4.6 are permitted to be used to determine the MCE_R response spectrum for the site of interest. The site-specific ground motion procedures set forth in Chapter 21 are also permitted to be used to determine ground motions for any isolated structure. For isolated structures on Site Class F sites, site response analysis shall be performed in accordance with Section 21.1.

17.3.2 MCE_R Response Spectra and Spectral Response Acceleration Parameters, S_{MS}, S_{M1}. The MCE_R response spectrum shall be the MCE_R response spectrum of Sections 11.4.6, or 11.4.7. The MCE_R response spectral acceleration parameters S_{MS} and S_{M1} shall be determined in accordance with Section 11.4.4 or 11.4.8.

17.3.3 MCE_R Ground Motion Records. Where response history analysis procedures are used, MCE_R ground motions shall consist of not less than seven pairs of horizontal acceleration components selected and scaled from individual recorded events that have magnitudes, fault distances, and source mechanisms that are consistent with those that control the MCE_R. Amplitude or spectral matching is permitted to scale the ground motions. Where the required number of recorded ground motion pairs is not available, simulated ground motion pairs are permitted to make up the total number required.

For each pair of horizontal ground motion components, a square root of the sum of the squares (SRSS) spectrum shall be constructed by taking the SRSS of the 5%-damped response spectra for the scaled components (when amplitude scaling is used, an identical scale factor is applied to both components of a pair). Each pair of motions shall be scaled such that in the period range from $0.75T_M$, determined using upper bound isolation system properties, to $1.25T_M$, determined using lower bound isolation system properties, the average of the SRSS spectra from all horizontal component pairs does not fall below the corresponding ordinate of the response spectrum used in the design (MCE_R), determined in accordance with Section 11.4.6 or 11.4.7.

For records that are spectrally matched, each pair of motions shall be scaled such that in the period range from $0.2T_M$, determined using upper bound properties, to $1.25T_M$, determined using lower bound properties, the response spectrum of one component of the pair is at least 90% of the corresponding ordinate of the response spectrum used in the design determined in accordance with Section 11.4.6 or 11.4.7.

For sites within 3 mi (5 km) of the active fault that controls the hazard, spectral matching shall not be used unless the pulse characteristics of the near-field ground motions are included in the site-specific response spectra, and pulse characteristics, when present in individual ground motions, are retained after the matching process has been completed.

At sites within 3 mi (5 km) of the active fault that controls the hazard, each pair of components shall be rotated to the fault-normal and fault-parallel directions of the causative fault and shall be scaled so that the average spectrum of the fault-normal components is not less than the MCE_R spectrum and the average spectrum of the fault-parallel components is not less than 50% of the MCE_R response spectrum for the period range $0.2T_M$, determined using upper bound properties, to $1.25T_M$, determined using lower bound properties.

17.4 ANALYSIS PROCEDURE SELECTION

Seismically isolated structures except those defined in Section 17.4.1 shall be designed using the dynamic procedures of Section 17.6. Where supplementary viscous dampers are used, the response history analysis procedures of Section 17.4.2.2 shall be used.

17.4.1 Equivalent Lateral Force Procedure. The equivalent lateral force procedure of Section 17.5 is permitted to be used for design of a seismically isolated structure provided that all of the following items are satisfied. These requirements shall be evaluated separately for upper bound and lower bound isolation system properties, and the more restrictive requirement shall govern.

1. The structure is located on a Site Class A, B, C, or D site.
2. The effective period of the isolated structure at the maximum displacement, D_M, is less than or equal to 5.0 s.
3. The structure above the isolation interface is less than or equal to four stories or 65 ft (19.8 m) in structural height measured from the base level.

   EXCEPTION: These limits are permitted to be exceeded if there is no tension or uplift on the isolators.
4. The effective damping of the isolation system at the maximum displacement, D_M, is less than or equal to 30%.
5. The effective period of the isolated structure T_M is greater than three times the elastic, fixed-base period of the structure above the isolation system, determined using a rational modal analysis.
6. The structure above the isolation system does not have a structural irregularity, as defined in Section 17.2.2.
7. The isolation system meets all of the following criteria:
   a. The effective stiffness of the isolation system at the maximum displacement is greater than one-third of the effective stiffness at 20% of the maximum displacement.
   b. The isolation system is capable of producing a restoring force, as specified in Section 17.2.4.4.
   c. The isolation system does not limit maximum earthquake displacement to less than the total maximum displacement, D_{TM}.

17.4.2 Dynamic Procedures. The dynamic procedures of Section 17.6 are permitted to be used as specified in this section.

17.4.2.1 Response Spectrum Analysis Procedure. Response spectrum analysis procedure shall not be used for design of a seismically isolated structure unless the structure, site, and isolation system meet the criteria of Section 17.4.1, Items 1, 2, 3, 4, and 6.

17.4.2.2 Response History Analysis Procedure. The response history analysis procedure is permitted to be used for design of any seismically isolated structure and shall be used for design of all seismically isolated structures not meeting the criteria of Section 17.4.2.1.

17.5 EQUIVALENT LATERAL FORCE PROCEDURE

17.5.1 General. Where the equivalent lateral force procedure is used to design seismically isolated structures, the requirements of this section shall apply.

17.5.2 Deformation Characteristics of the Isolation System. Minimum lateral earthquake design displacements and forces on seismically isolated structures shall be based on the deformation characteristics of the isolation system. The deformation characteristics of the isolation system include the effects of the wind-restraint system if such a system is used to meet the design requirements of this standard. The deformation characteristics of the isolation system shall be based on properly substantiated prototype tests performed in accordance with Section 17.8 and shall incorporate property modification factors in accordance with Section 17.2.8.4.

The analysis of the isolation system and structure shall be performed separately for upper bound and lower bound properties, and the governing case for each response parameter of interest shall be used for design.

17.5.3 Minimum Lateral Displacements Required for Design

17.5.3.1 Maximum Displacement. The isolation system shall be designed and constructed to withstand, at a minimum, the maximum displacement, D_M, determined using upper bound and lower bound properties, in the most critical direction of horizontal response, calculated using Eq. (17.5-1):

$$D_M = \frac{g S_{M1} T_M}{4\pi^2 B_M} \tag{17.5-1}$$

where

g = acceleration caused by gravity [in./s^2 (mm/s^2)] if the units of the displacement D_M are in. (mm);

Table 17.5-1 Damping Factor, B_M

Effective Damping, β_M (percentage of critical)[a,b]	B_M Factor
≤2	0.8
5	1.0
10	1.2
20	1.5
30	1.7
40	1.9
≥50	2.0

[a]The damping factor shall be based on the effective damping of the isolation system determined in accordance with the requirements of Section 17.2.8.6.
[b]The damping factor shall be based on linear interpolation for effective damping values other than those given.

S_{M1} = MCE$_R$ 5% damped spectral acceleration parameter at 1-s period in units of g-sec, as determined in Section 11.4.4 or 11.4.8;
T_M = effective period of the seismically isolated structure [s] at the displacement D_M in the direction under consideration, as prescribed by Eq. (17.5-2); and
B_M = numerical coefficient as set forth in Table 17.5-1 for the effective damping of the isolation system β_M at the displacement D_M.

17.5.3.2 Effective Period at the Maximum Displacement. The effective period of the isolated structure, T_M, at the maximum displacement, D_M, shall be determined using upper bound and lower bound deformational characteristics of the isolation system and Eq. (17.5-2):

$$T_M = 2\pi\sqrt{\frac{W}{k_M g}} \tag{17.5-2}$$

where

W = effective seismic weight of the structure above the isolation interface as defined in Section 12.7.2;
k_M = effective stiffness [kip/in. (kN/mm)] of the isolation system at the maximum displacement, D_M, as prescribed by Eq. (17.2-3); and
g = acceleration caused by gravity [in./s^2 (mm/s^2)] if the units of k_M are in kip/in. (kN/mm).

17.5.3.3 Total Maximum Displacement. The total maximum displacement, D_{TM}, of elements of the isolation system shall include additional displacement caused by actual and accidental torsion calculated from the spatial distribution of the lateral stiffness of the isolation system and the most disadvantageous location of eccentric mass. The total maximum displacement, D_{TM}, of elements of an isolation system shall not be taken as less than that prescribed by Eq. (17.5-3):

$$D_{TM} = D_M\left[1 + \left(\frac{y}{P_T^2}\right)\frac{12e}{b^2 + d^2}\right] \tag{17.5-3}$$

where

D_M = displacement at the center of rigidity of the isolation system in the direction under consideration as prescribed by Eq. (17.5-1);
y = the distance [in. (mm)] between the centers of rigidity of the isolation system and the element of interest measured perpendicular to the direction of seismic loading under consideration;
e = the actual eccentricity measured in plan between the center of mass of the structure above the isolation interface and the center of rigidity of the isolation system, plus accidental eccentricity [ft (mm)], taken as 5% of the longest plan dimension of the structure perpendicular to the direction of force under consideration;
b = the shortest plan dimension of the structure [ft (mm)] measured perpendicular to d;
d = the longest plan dimension of the structure [ft (mm)]; and
P_T = ratio of the effective translational period of the isolation system to the effective torsional period of the isolation system, as calculated by dynamic analysis or as prescribed by Eq. (17.5-4) but need not be taken as less than 1.0.

$$P_T = \frac{1}{r_I}\sqrt{\frac{\sum_{i=1}^{N}(x_i^2 + y_i^2)}{N}} \tag{17.5-4}$$

where

x_i, y_i = horizontal distances [ft (mm)] from the center of mass to the ith isolator unit in the two horizontal axes of the isolation system;
N = number of isolator units;
r_I = radius of gyration of the isolation system [ft (mm)], which is equal to $((b^2 + d^2)/12)^{1/2}$ for isolation systems of rectangular plan dimension, $b \times d$.

The total maximum displacement, D_{TM}, shall not be taken as less than 1.15 times D_M.

17.5.4 Minimum Lateral Forces Required for Design

17.5.4.1 Isolation System and Structural Elements below the Base Level. The isolation system, the foundation, and all structural elements below the base level shall be designed and constructed to withstand a minimum lateral seismic force, V_b, using all of the applicable requirements for a nonisolated structure as prescribed by the value of Eq. (17.5-5), determined using both upper bound and lower bound isolation system properties:

$$V_b = k_M D_M \tag{17.5-5}$$

where

k_M = effective stiffness [kip/in. (kN/mm)] of the isolation system at the displacement D_M, as prescribed by Eq. (17.2-3) and
D_M = maximum displacement [in. (mm)] at the center of rigidity of the isolation system in the direction under consideration, as prescribed by Eq. (17.5-1).

V_b shall not be taken as less than the maximum force in the isolation system at any displacement up to and including the maximum displacement D_M, as defined in Section 17.5.3

Overturning loads on elements of the isolation system, the foundation, and structural elements below the base level caused by lateral seismic force V_b shall be based on the vertical distribution of force of Section 17.5.5, except that the unreduced lateral seismic design force V_{st} shall be used in lieu of V_s in Eq. (17.5-9).

17.5.4.2 Structural Elements above the Base Level. The structure above the base level shall be designed and constructed using all of the applicable requirements for a non-isolated structure for a minimum shear force, V_s, determined using upper bound and lower bound isolation system properties, as prescribed by Eq. (17.5-6):

$$V_s = \frac{V_{st}}{R_I} \tag{17.5-6}$$

where

R_I = numerical coefficient related to the type of seismic force-resisting system above the isolation system; and
V_{st} = total unreduced lateral seismic design force or shear on elements above the base level, as prescribed by Eq. (17.5-7).

The R_I factor shall be based on the type of seismic force-resisting system used for the structure above the base level in the direction of interest and shall be three-eighths of the value of R given in Table 12.2-1, with a maximum value not greater than 2.0 and a minimum value not less than 1.0.

EXCEPTION: The value of R_i is permitted to be taken as greater than 2.0, provided the strength of the structure above the base level in the direction of interest, as determined by nonlinear static analysis at a roof displacement corresponding to a maximum story drift the lesser of the MCE$_R$ drift or $0.015h_{sx}$, is not less than 1.1 times V_b.

The total unreduced lateral seismic force or shear on elements above the base level shall be determined using upper bound and lower bound isolation system properties, as prescribed by Eq. (17.5-7):

$$V_{st} = V_b \left(\frac{W_s}{W}\right)^{(1-2.5\beta m)} \tag{17.5-7}$$

where

W = effective seismic weight [kips (kN)] of the structure above the isolation interface as defined in Section 12.7.2; and
W_s = effective seismic weight [kips (kN)] of the structure above the isolation interface as defined in Section 12.7.2, excluding the effective seismic weight [kips (kN)] of the base level.

The effective seismic weight W_s in Eq. (17.5-7) shall be taken as equal to W when the average distance from the top of the isolator to the underside of the base level floor framing above the isolators exceeds 3 ft (0.9 m).

EXCEPTION: For isolation systems whose hysteretic behavior is characterized by an abrupt transition from preyield to postyield or preslip to postslip behavior, the exponent term (1–2.5β_M) in Eq. (17.5-7) shall be replaced by (1–3.5β_M).

17.5.4.3 Limits on V_s. The value of V_s shall not be taken as less than each of the following:

1. The lateral seismic force required by Section 12.8 for a fixed-base structure of the same effective seismic weight, W_s, and a period equal to the period of the isolation system using the upper bound properties T_M;
2. The base shear corresponding to the factored design wind load; and
3. The lateral seismic force, V_{st}, calculated using Eq. (17.5-7), and with V_b set equal to the force required to fully activate the isolation system using the greater of the upper bound properties, or
   a. 1.5 times the nominal properties for the yield level of a softening system,
   b. the ultimate capacity of a sacrificial wind-restraint system,
   c. the breakaway friction force of a sliding system, or
   d. the force at zero displacement of a sliding system following a complete dynamic cycle of motion at D_M.

17.5.5 Vertical Distribution of Force. The lateral seismic force V_s shall be distributed over the height of the structure above the base level, using upper bound and lower bound isolation system properties, using the following equations:

$$F_1 = \frac{(V_b - V_{st})}{R_I} \tag{17.5-8}$$

and

$$F_x = C_{vx} V_s \tag{17.5-9}$$

and

$$C_{vx} = \frac{w_x h_x^k}{\sum_{i=2}^{n} w_i h_i^k} \tag{17.5-10}$$

and

$$k = 14\beta_M T_{fb} \tag{17.5-11}$$

where

F_1 = lateral seismic force [kips (kN)] induced at level 1, the base level;
F_x = lateral seismic force [kips (kN)] induced at level x, $x > 1$;
C_{vx} = vertical distribution factor;
V_s = total lateral seismic design force or shear on elements above the base level as prescribed by Eq. (17.5-6) and the limits of Section 17.5.4.3;
$w_i w_x$ = portion of W_s that is located at or assigned to level i or x;
$h_i h_x$ = height above the isolation interface of level i or x; and
T_{fb} = the fundamental period, in s, of the structure above the isolation interface determined using a rational modal analysis assuming fixed-base conditions.

EXCEPTION: In lieu of Eqs. (17.5-6) and (17.5-9), the lateral seismic force F_x is permitted to be calculated as the average value of the force at level x in the direction of interest using the results of a simplified stick model of the building and a lumped representation of the isolation system using response history analysis scaled to V_b/R_I at the base level.

17.5.6 Drift Limits. The maximum story drift of the structure above the isolation system shall not exceed $0.015h_{sx}$. The drift shall be calculated by Eq. (12.8-15) with C_d for the isolated structure equal to R_I as defined in Section 17.5.4.2.

17.6 DYNAMIC ANALYSIS PROCEDURES

17.6.1 General. Where dynamic analysis is used to design seismically isolated structures, the requirements of this section shall apply.

17.6.2 Modeling. The mathematical models of the isolated structure, including the isolation system, the seismic force-resisting system, and other structural elements, shall conform to Section 12.7.3 and to the requirements of Sections 17.6.2.1 and 17.6.2.2.

17.6.2.1 Isolation System. The isolation system shall be modeled using deformational characteristics developed in accordance with Section 17.2.8. The lateral displacements and forces shall be computed separately for upper bound and lower bound isolation system properties as defined in Section 17.2.8.5. The isolation system shall be modeled with sufficient detail to capture all of the following:

1. Spatial distribution of isolator units.
2. Translation, in both horizontal directions, and torsion of the structure above the isolation interface considering the most disadvantageous location of eccentric mass.
3. Overturning and uplift forces on individual isolator units.
4. Effects of vertical load, bilateral load, and/or the rate of loading if the force-deflection properties of the isolation system are dependent on one or more of these attributes.

The total maximum displacement, D_{TM}, across the isolation system shall be calculated using a model of the isolated structure that incorporates the force-deflection characteristics of nonlinear elements of the isolation system and the seismic force-resisting system.

17.6.2.2 Isolated Structure. The maximum displacement of each floor and design forces and displacements in elements of the seismic force-resisting system are permitted to be calculated using a linear elastic model of the isolated structure provided that all elements of the seismic force-resisting system of the structure above the isolation system remain essentially elastic.

Seismic force-resisting systems with essentially elastic elements include, but are not limited to, regular structural systems designed for a lateral force not less than 100% of V_s determined in accordance with Sections 17.5.4.2 and 17.5.4.3.

The analysis of the isolation system and structure shall be performed separately for upper bound and lower bound properties, and the governing case for each response parameter of interest shall be used for design.

17.6.3 Description of Procedures

17.6.3.1 General. Response spectrum analysis shall be performed in accordance with Section 12.9 and the requirements of Section 17.6.3.3. Response history analysis shall be performed in accordance with the requirements of Section 17.6.3.4.

17.6.3.2 MCE$_R$ Ground Motions. The MCE$_R$ ground motions of Section 17.3 shall be used to calculate the lateral forces and displacements in the isolated structure, the total maximum displacement of the isolation system, and the forces in the isolator units, isolator unit connections, and supporting framing immediately above and below the isolators used to resist isolator P-delta demands.

17.6.3.3 Response Spectrum Analysis Procedure. Response spectrum analysis shall be performed using a modal damping value for the fundamental mode in the direction of interest not greater than the effective damping of the isolation system or 30% of critical, whichever is less. Modal damping values for higher modes shall be selected consistent with those that would be appropriate for response spectrum analysis of the structure above the isolation system assuming a fixed base.

Response spectrum analysis used to determine the total maximum displacement shall include simultaneous excitation of the model by 100% of the ground motion in the critical direction and 30% of the ground motion in the perpendicular, horizontal direction. The maximum displacement of the isolation system shall be calculated as the vector sum of the two orthogonal displacements.

17.6.3.4 Response History Analysis Procedure. Response history analysis shall be performed for a set of ground motion pairs selected and scaled in accordance with Section 17.3.3. Each pair of ground motion components shall be applied simultaneously to the model, considering the most disadvantageous location of eccentric mass. The maximum displacement of the isolation system shall be calculated from the vector sum of the two orthogonal displacements at each time step.

The parameters of interest shall be calculated for each ground motion used for the response history analysis, and the average value of the response parameter of interest shall be used for design.

For sites identified as near-fault, each pair of horizontal ground motion components shall be rotated to the fault-normal and fault-parallel directions of the causative faults and applied to the building in such orientation.

For all other sites, individual pairs of horizontal ground motion components need not be applied in multiple orientations.

17.6.3.4.1 Accidental Mass Eccentricity. Torsional response resulting from lack of symmetry in mass and stiffness shall be accounted for in the analysis. In addition, accidental eccentricity consisting of displacement of the center of mass from the computed location by an amount equal to 5% of the diaphragm dimension, separately in each of two orthogonal directions at the level under consideration.

The effects of accidental eccentricity are permitted to be accounted for by amplifying forces, drifts, and deformations determined from an analysis using only the computed center of mass, provided that factors used to amplify forces, drifts, and deformations of the center-of-mass analysis are shown to produce results that bound all the mass-eccentric cases.

17.6.4 Minimum Lateral Displacements and Forces

17.6.4.1 Isolation System and Structural Elements below the Base Level. The isolation system, foundation, and all structural elements below the base level shall be designed using all of the applicable requirements for a nonisolated structure and the forces obtained from the dynamic analysis without reduction, but the design lateral force shall not be taken as less than 90% of V_b determined by Eq. (17.5-5).

The total maximum displacement of the isolation system shall not be taken as less than 80% of D_{TM} as prescribed by Section 17.5.3.3 except that D'_M is permitted to be used in lieu of D_M where

$$D'_M = \frac{D_M}{\sqrt{1 + (T/T_M)^2}} \tag{17.6-1}$$

and

D_M = maximum displacement [in. (mm)] at the center of rigidity of the isolation system in the direction under consideration, as prescribed by Eq. (17.5-1);

T = elastic, fixed-base period, in s, of the structure above the isolation system as determined by Section 12.8.2, including the coefficient C_u, if the approximate period formulas are used to calculate the fundamental period; and

T_M = effective period, in s, of the seismically isolated structure, at the displacement D_M in the direction under consideration, as prescribed by Eq. (17.5-2).

17.6.4.2 Structural Elements above the Base Level. Subject to the procedure-specific limits of this section, structural elements above the base level shall be designed using the applicable requirements for a nonisolated structure and the forces obtained from the dynamic analysis reduced by a factor of R_I as determined in accordance with Section 17.5.4.2.

For response spectrum analysis, the design shear at any story shall not be less than the story shear resulting from application of the forces calculated using Eq. (17.5-9) and a value of V_b equal to the base shear obtained from the response spectrum analysis in the direction of interest.

For response history analysis of regular structures, the value of V_b shall not be taken as less than 80% of that determined in accordance with Section 17.5.4.1, and the value V_s shall not be taken as less than 100% of the limits specified by Section 17.5.4.3.

For response history analysis of irregular structures, the value of V_b shall not be taken as less than 100% of that determined in accordance with Section 17.5.4.1, and the value V_s shall not be taken as less than 100% of the limits specified by Section 17.5.4.3.

17.6.4.3 Scaling of Results. Where the factored lateral shear force on structural elements, determined using either the response spectrum or response history procedure, is less than the minimum values prescribed by Sections 17.6.4.1 and 17.6.4.2, all design parameters shall be adjusted upward proportionally.

17.6.4.4 Drift Limits. Maximum story drift corresponding to the design lateral force including displacement caused by vertical deformation of the isolation system shall comply with either of the following limits:

1. Where response spectrum analysis is used, the maximum story drift of the structure above the isolation system shall not exceed $0.015h_{sx}$.
2. Where response history analysis based on the force-deflection characteristics of nonlinear elements of the seismic force-resisting system is used, the maximum story drift of the structure above the isolation system shall not exceed $0.020h_{sx}$.

Drift shall be calculated using Eq. (12.8-15) with the C_d of the isolated structure equal to R_I as defined in Section 17.5.4.2.

The secondary effects of the maximum lateral displacement of the structure above the isolation system combined with gravity forces shall be investigated if the story drift ratio exceeds $0.010/R_I$.

17.7 DESIGN REVIEW

An independent design review of the isolation system and related test programs shall be performed by one or more individuals possessing knowledge of the following items with a minimum of one reviewer being a registered design professional (RDP). Isolation system design review shall include, but not be limited to, all of the following:

1. Project design criteria, including site-specific spectra and ground motion histories.
2. Preliminary design, including the selection of the devices, determination of the maximum displacement, the total maximum displacement, and the lateral force level.
3. Review of qualification data and appropriate property modification factors for the manufacturer and device selected.
4. Prototype testing program (Section 17.8.2).
5. Final design of the entire structural system and all supporting analyses, including modeling of isolators for response history analysis if performed.
6. Isolator production testing program (Section 17.8.5).

17.8 TESTING

17.8.1 General. The deformation characteristics and damping values of the isolation system used in the design and analysis of seismically isolated structures shall be based on tests of a selected sample of the components before construction as described in this section. The isolation system components to be tested shall include the wind-restraint system if such a system is used in the design.

The tests specified in this section are for establishing and validating the isolator unit and isolation system test properties that are used to determine design properties of the isolation system in accordance with Section 17.2.8

17.8.1.1 Qualification Tests. Isolation device manufacturers shall submit for approval by the registered design professional the results of qualification tests, analysis of test data, and supporting scientific studies that are permitted to be used to quantify the effects of heating caused by cyclic dynamic motion, loading rate, scragging, variability and uncertainty in production bearing properties, temperature, aging, environmental exposure, and contamination. The qualification testing shall be applicable to the component types, models, materials, and sizes to be used in the construction. The qualification testing shall have been performed on components manufactured by the same manufacturer supplying the components to be used in the construction. When scaled specimens are used in the qualification testing, principles of scaling and similarity shall be used in the interpretation of the data.

17.8.2 Prototype Tests. Prototype tests shall be performed separately on two full-size specimens (or sets of specimens, as appropriate) of each predominant type and size of isolator unit of the isolation system. The test specimens shall include the wind-restraint system if such a system is used in the design. Specimens tested shall not be used for construction unless they are accepted by the registered design professional responsible for the design of the structure.

17.8.2.1 Record. For each cycle of each test, the force-deflection behavior of the test specimen shall be recorded.

17.8.2.2 Sequence and Cycles. Each of the following sequence of tests shall be performed for the prescribed number of cycles at a vertical load equal to the average dead load plus one-half the effects caused by live load on all isolator units of a common type and size. Before these tests, the production set of tests specified in Section 17.8.5 shall be performed on each isolator:

1. Twenty fully reversed cycles of loading at a lateral force corresponding to the wind design force.
2. The sequence of either item (a) or item (b) below shall be performed:
   a. Three fully reversed cycles of loading at each of the following increments of the displacement: $0.25D_M$, $0.5D_M$, $0.67D_M$, and $1.0D_M$ where D_M is determined in Section 17.5.3.1 or Section 17.6, as appropriate.
   b. The following sequence, performed dynamically at the effective period, T_M: continuous loading of one fully reversed cycle at each of the following increments of the maximum displacement: $1.0D_M$, $0.67D_M$, $0.5D_M$, and $0.25D_M$ followed by continuous loading of one fully reversed cycle at $0.25D_M$, $0.5D_M$, $0.67D_M$, and $1.0D_M$. A rest interval is permitted between these two sequences.
3. Three fully reversed cycles of loading at the maximum displacement, $1.0D_M$.
4. The sequence of either item (a) or item (b) below shall be performed:
   a. $30S_{M1}/(S_{MS}B_M)$, but not fewer than 10, continuous fully reversed cycles of loading at 0.75 times the maximum displacement, $0.75D_M$.
   b. The test of item (a), performed dynamically at the effective period, T_M. This test may comprise separate sets of multiple cycles of loading, with each set consisting of not fewer than five continuous cycles.

If an isolator unit is also a vertical load-carrying element, then item 3 of the sequence of cyclic tests specified in the preceding text shall be performed for two additional vertical load cases specified in Section 17.2.7.1. The load increment caused by earthquake overturning, Q_E, shall be equal to or greater than the peak earthquake vertical force response corresponding to the test displacement being evaluated. In these tests, the combined vertical load shall be taken as the typical or average downward force on all isolator units of a common type and size. Axial load and displacement values for each test shall be the greater of those determined by analysis using upper bound and lower bound values of isolation system properties determined in accordance with Section 17.2.8.5. The effective period T_M shall be the lower of those determined by analysis using upper bound and lower bound values.

17.8.2.3 Dynamic Testing. Tests specified in Section 17.8.2.2 shall be performed dynamically at the lower of the effective periods, T_M, determined using upper bound and lower bound properties.

Dynamic testing shall not be required if the prototype testing has been performed dynamically on similar-sized isolators that meet the requirements of Section 17.8.2.7, and the testing was conducted at similar loads and accounted for the effects of velocity, amplitude of displacement, and heating effects. The prior dynamic prototype test data shall be used to establish factors that adjust three-cycle average values of k_d and E_{loop} to account for the difference in test velocity and heating effects and to establish $\lambda_{(\text{test, min})}$ and $\lambda_{(\text{test, max})}$.

Only if full-scale testing is not possible, reduced-scale prototype specimens can be used to quantify rate-dependent properties of isolators. The reduced-scale prototype specimens shall be of the same type and material and shall be manufactured with the same processes and quality as full-scale prototypes and shall be tested at a frequency that represents full-scale prototype loading rates.

17.8.2.4 Units Dependent on Bilateral Load. If the force-deflection properties of the isolator units exhibit bilateral load dependence, the tests specified in Sections 17.8.2.2 and 17.8.2.3 shall be augmented to include bilateral load at the following increments of the maximum displacement, D_M: 0.25 and 1.0, 0.5 and 1.0, 0.67 and 1.0, and 1.0 and 1.0.

If reduced-scale specimens are used to quantify bilateral-load-dependent properties, they shall meet the requirements of Section 17.8.2.7; the reduced-scale specimens shall be of the same type and material and manufactured with the same processes and quality as full-scale prototypes.

The force-deflection properties of an isolator unit shall be considered to be dependent on bilateral load if the effective stiffness when subjected to bilateral loading is different by more than 15% from the effective stiffness subjected to unilateral loading.

17.8.2.5 Maximum and Minimum Vertical Load. Isolator units that carry vertical load shall be subjected to one fully reversed cycle of loading at the total maximum displacement, D_{TM}, and at each of the vertical loads corresponding to the maximum and minimum downward vertical loads as specified in Section 17.2.7.1 on any one isolator of a common type and size. Axial load and displacement values for each test shall be the greater of those determined by analysis using the upper bound and lower bound values of isolation system properties determined in accordance with Section 17.2.8.5.

EXCEPTION: In lieu of envelope values for a single test, it shall be acceptable to perform two tests, one each for the combination of vertical load and horizontal displacement obtained from analysis using the upper bound and lower bound values of isolation system properties, respectively, determined in accordance with Section 17.2.8.5

17.8.2.6 Sacrificial Wind-Restraint Systems. If a sacrificial wind-restraint system is to be used, its ultimate capacity shall be established by test.

17.8.2.7 Testing Similar Units. Prototype tests need not be performed if an isolator unit, when compared to another tested unit, complies with all of the following criteria:

1. The isolator design is not more than 15% larger nor more than 30% smaller than the previously tested prototype, in terms of governing device dimensions; and
2. The design is of the same type and materials; and
3. The design has an energy dissipated per cycle, E_{loop}, that is not less than 85% of the previously tested unit; and
4. The design is fabricated by the same manufacturer using the same or more stringent documented manufacturing and quality control procedures; and
5. For elastomeric type isolators, the design shall not be subject to a greater shear strain nor greater vertical stress than that of the previously tested prototype; and
6. For sliding type isolators, the design shall not be subject to a greater vertical stress or sliding velocity than that of the previously tested prototype using the same sliding material.

The prototype testing exemption above shall be approved by independent design review, as specified in Section 17.7.

When the results of tests of similar isolator units are used to establish dynamic properties in accordance with Section 17.8.2.3, in addition to Items 2 to 4 above, the following criteria shall be satisfied:

7. The similar unit shall be tested at a frequency that represents design full-scale loading rates in accordance with principles of scaling and similarity.
8. The length scale of reduced-scale specimens shall not be greater than two.

17.8.3 Determination of Force-Deflection Characteristics. The force-deflection characteristics of an isolator unit shall be based on the cyclic load tests of prototype isolators specified in Section 17.8.2.

As required, the effective stiffness of an isolator unit, k_{eff}, shall be calculated for each cycle of loading as prescribed by Eq. (17.8-1):

$$k_{\mathrm{eff}} = \frac{|F^+| + |F^-|}{|\Delta^+| + |\Delta^-|} \quad (17.8\text{-}1)$$

where F^+ and F^- are the positive and negative forces, at the maximum positive and minimum negative displacements Δ^+ and Δ^-, respectively.

As required, the effective damping, β_{eff}, of an isolator unit shall be calculated for each cycle of loading by Eq. (17.8-2):

$$\beta_{\mathrm{eff}} = \frac{2}{\pi} \frac{E_{\mathrm{loop}}}{k_{\mathrm{eff}}(|\Delta^+| + |\Delta^-|)^2} \quad (17.8\text{-}2)$$

where the energy dissipated per cycle of loading, E_{loop}, and the effective stiffness, k_{eff}, shall be based on peak test displacements of Δ^+ and Δ^-.

As required, the postyield stiffness, k_d, of each isolator unit shall be calculated for each cycle of loading using the following assumptions:

1. A test loop shall be assumed to have bilinear hysteretic characteristics with values of k_1, k_d, f_o, f_y, k_{eff}, and E_{loop} as shown in Fig. 17.8-1.
2. The computed loop shall have the same values of effective stiffness, k_{eff}, and energy dissipated per cycle of loading, E_{loop}, as the test loop.

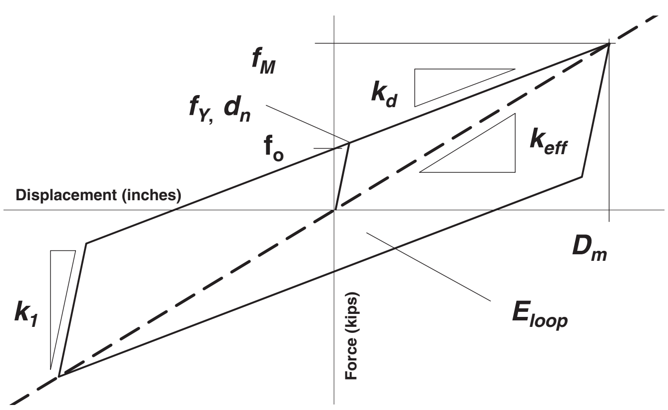

FIGURE 17.8-1 Nominal Properties of the Isolator Bilinear Force-Deflection Model

3. The assumed value of k_1 shall be a visual fit to the elastic stiffness of the isolator unit during unloading immediately after D_M.

It is permitted to use different methods for fitting the loop, such as a straight-line fit of k_d directly to the hysteresis curve extending to D_M and then determining k_1 to match E_{loop}.

17.8.4 Test Specimen Adequacy. The performance of the test specimens shall be deemed adequate if all of the following conditions are satisfied:

1. The force-deflection plots for all tests specified in Section 17.8.2 have a positive incremental force-resisting capacity.
2. The average postyield stiffness, k_d, and energy dissipated per cycle, E_{loop}, for the three cycles of test specified in Section 17.8.2.2, Item 3, for the vertical load equal to the average dead load plus one-half the effects caused by live load, including the effects of heating and rate of loading in accordance with Section 17.2.8.3, shall fall within the range of the nominal design values defined by the permissible individual isolator range, which are typically +/–5% greater than the $\lambda_{(\mathrm{spec,\,min})}$ and $\lambda_{(\mathrm{spec,\,max})}$ range for the average of all isolators.
3. For each increment of test displacements $0.67D_M$ and $1.0D_M$ specified in Item 2 and Item 3 of Section 17.8.2.2 and for each vertical load case specified in Section 17.8.2.2, the value of the postyield stiffness, k_d, at each of the cycles of test at a common displacement shall fall within the range defined by $\lambda_{(\mathrm{test,\,min})}$ and $\lambda_{(\mathrm{test,\,max})}$ multiplied by the nominal value of postyield stiffness.
4. For each specimen, there is no greater than a 20% change in the initial effective stiffness over the cycles of test specified in Item 4 of Section 17.8.2.2.
5. For each test specimen, the value of the postyield stiffness, k_d, and energy dissipated per cycle, E_{loop}, for any cycle of test in Section 17.8.2.2, Item 4(a) shall fall within the range of the nominal design values defined by $\lambda_{(\mathrm{test,\,min})}$ and $\lambda_{(\mathrm{test,\,max})}$.
6. For each specimen, there is no greater than a 20% decrease in the initial effective damping over the cycles of test specified in Item 4 of Section 17.8.2.2.
7. All specimens of vertical load-carrying elements of the isolation system remain stable where tested in accordance with Section 17.8.2.5.

EXCEPTION: The registered design professional is permitted to adjust the limits of Items 3, 4, and 6 to account for the property variation factors of Section 17.2.8.4 used for design of the isolation system.

17.8.5 Production Tests. A test program for the isolator units used in the construction shall be established by the registered design professional. The test program shall evaluate the consistency of measured values of nominal isolator unit properties by testing 100% of the isolators in combined compression and shear at not less than two-thirds of the maximum displacement, D_M, determined using lower bound properties.

The mean results of all tests shall fall within the range of values defined by the $\lambda_{(\mathrm{spec,\,max})}$ and $\lambda_{(\mathrm{spec,\,min})}$ values established in Section 17.2.8.4. A different range of values is permitted to be used for individual isolator units and for the average value across all isolators of a given unit type, provided that differences in the ranges of values are accounted for in the design of each element of the isolation system, as prescribed in Section 17.2.8.4.

17.9 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

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